

Sai Madhu Sudhan Reddy

Software Engineer — Backend & Enterprise Systems

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SUMMARY

Software Engineer with 2 years of production backend experience building and operating enterprise applications used by 1,000+ users. Core strength is designing reliable PostgreSQL/Oracle-backed services in Python and Java, owning incident response, and shipping features across the full stack. Outside of work, I design and build AI systems end to end — LLM applications, retrieval pipelines, and secure AI infrastructure — as self-initiated projects, including a nationally recognized entry (Top 10, Google Build with Gemma) and two IEEE conference publications.

CORE COMPETENCIES

Languages: Python, Java, SQL, JavaScript/TypeScript

Backend & APIs: FastAPI, Spring Boot, REST API design, JWT/RBAC auth

Databases: PostgreSQL, Oracle, MySQL, DuckDB — schema design, query tuning, performance analysis

DevOps & Quality: Docker, Nginx, Git, SonarQube static analysis, code review

AI / ML (personal projects): LLM application design (Llama 3.1, Gemma), RAG pipelines, Qdrant vector search, prompt & context engineering

PROFESSIONAL EXPERIENCE

Software Engineer — Kallque Pvt. Ltd.

Apr 2026 – Present

- Architected and maintained production Python/SQL backend services supporting 1,000+ users across the PG Owner and PG Tenant platforms.
- Owned root-cause analysis and resolution of 8–15 production incidents per month, improving application stability and reducing recurring issues.
- Led database performance analysis across 13+ production tables, identifying and resolving query bottlenecks to improve reliability.
- Partnered with product, QA, and frontend teams to ship production-ready features and database enhancements on schedule.

Trainee Software Engineer — Kallque Pvt. Ltd.

Dec 2024 – Mar 2026

- Resolved 100+ application and database defects across production and testing environments.
- Built validation logic and DTO enhancements in a Java/Spring Boot platform supporting tenant, payment, and occupancy management workflows.
- Raised team code quality through 200+ code reviews and SonarQube-driven static analysis.
- Optimized SQL queries and supported database debugging, improving application reliability and maintainability.

Software Engineering Intern — Kallque Pvt. Ltd.

Jun 2024 – Nov 2024

- Built foundational experience in enterprise backend development, PostgreSQL, and defect investigation.

PERSONAL PROJECTS — AI SYSTEMS (SELF-INITIATED, OUTSIDE OF WORK)

Independent projects designed, built, and shipped on my own time — not part of any employer's codebase or product.

Enterprise AI Database Assistant — *Natural-language querying platform across 3 database engines*

- Built a schema-aware NL-to-SQL pipeline (28 REST APIs, 18 UI routes) reaching 91% first-attempt accuracy, with a self-healing retry loop recovering 80% of failed queries.
- Designed LLM-driven BI features — KPI generation, insights, and chart recommendations — at 95% recommendation accuracy.
- Secured the platform with JWT/RBAC and AST-based SQL-injection defense, blocking 100% of 12 tested attack patterns across 34 automated tests.

RepoMind AI — *AI-powered code intelligence platform*

- Built AST-based parsing and semantic chunking for repository-wide code understanding.
- Engineered a multi-stage retrieval pipeline (Qdrant vector search + lexical reranking + context compression) with file/line-level citations.
- Delivered real-time streamed responses via SSE, secured with JWT/bcrypt and sliding-window rate limiting.

CHAKRA — **Enterprise AI Compliance Investigation Platform** — *Top 10, Google Build with Gemma (national AI hackathon)*

- Designed an AI-assisted investigation platform to detect invoice anomalies, GST inconsistencies, and financial risk for SMEs, with OCR-assisted document processing.
- Built an explainable AI compliance workflow combining Gemma, DuckDB, a rule engine, and a knowledge graph; selected as a Top 10 team nationally.

Real-Time Heart Rate Estimation from Facial Video — *Published, IEEE*

- Built a computer-vision system (remote photoplethysmography) estimating heart rate from facial video across 200+ samples, at 68–80% accuracy.

PUBLICATIONS

- **Real-Time Estimation of Heart Rate from Facial Video** — IEEE
- **AI Methodologies in Upcoming Modern Warfare Systems** — IEEE

HONORS & RECOGNITION

- Top 10 Finalist — Google Build with Gemma, Bengaluru AI Sprint (national-level AI hackathon)
- Author, 2 IEEE conference publications

EDUCATION

B.Tech, Artificial Intelligence & Machine Learning — M.S. Ramaiah University of Applied Sciences

2024 | CGPA: 8.0 / 10